

8.4 Matrices for General Linear Transformations

In this section we show that a general linear transformation from any n -dimensional vector space V to any m -dimensional vector space W can be performed using an appropriate matrix transformation from R^n to R^m . This idea is used in computer computations since computers are well suited for performing matrix computations.

Matrices of Linear Transformations

Suppose V is n -dimensional, W is m -dimensional, and $T : V \rightarrow W$ is a linear transformation. Let $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis for V and $B' = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ a basis for W . For each $\mathbf{x} \in V$, the coordinate vectors $[\mathbf{x}]_B \in R^n$ and $[T(\mathbf{x})]_{B'} \in R^m$ are related by the following commutative diagram:

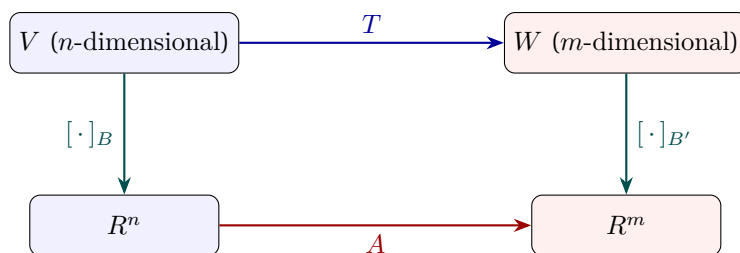


FIGURE 8.4.1

Our goal is to find an $m \times n$ matrix A such that

$$A[\mathbf{x}]_B = [T(\mathbf{x})]_{B'} \quad (1)$$

for all $\mathbf{x} \in V$. With such an A , computing $T(\mathbf{x})$ can be done *indirectly* in three steps:

Finding $T(\mathbf{x})$ Indirectly

- Step 1.** Compute the coordinate vector $[\mathbf{x}]_B$.
- Step 2.** Multiply $[\mathbf{x}]_B$ on the left by A to produce $[T(\mathbf{x})]_{B'}$.
- Step 3.** Reconstruct $T(\mathbf{x})$ from its coordinate vector $[T(\mathbf{x})]_{B'}$.

To find A , note that Equation (1) must hold for each basis vector $\mathbf{u}_i$ in B . Since $[\mathbf{u}_i]_B = \mathbf{e}_i$ (the i -th standard basis vector in R^n), multiplying A by $\mathbf{e}_i$ picks out the i -th column of A , which must equal $[T(\mathbf{u}_i)]_{B'}$. Thus the successive columns of A are the coordinate vectors of $T(\mathbf{u}_1), T(\mathbf{u}_2), \dots, T(\mathbf{u}_n)$ with respect to B' . We call this matrix the **matrix for T relative to the bases B and B'** , denoted $[T]_{B',B}$:

$$[T]_{B',B} = \left[[T(\mathbf{u}_1)]_{B'} \mid [T(\mathbf{u}_2)]_{B'} \mid \cdots \mid [T(\mathbf{u}_n)]_{B'} \right] \quad (4)$$

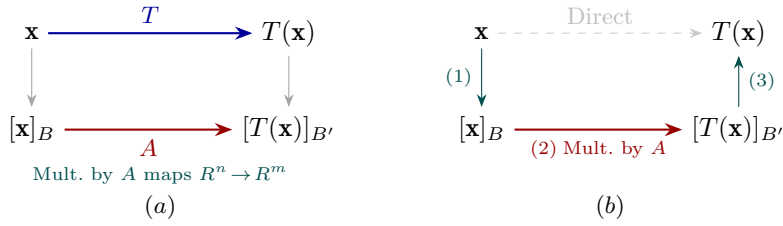


FIGURE 8.4.2

and this matrix satisfies

$$\boxed{[T]_{B',B}[\mathbf{x}]_B = [T(\mathbf{x})]_{B'}} \quad (5)$$

Remark. In the notation $[T]_{B',B}$, the *right* subscript B is the basis for the *domain* of T , and the *left* subscript B' is the basis for the *image space*. Observe how B "cancels out" in Formula (5): $[T]_{B',B}[\mathbf{x}]_B = [T(\mathbf{x})]_{B'}$.

When $T_C : R^n \rightarrow R^m$ is multiplication by C and S, S' are the standard bases for R^n and R^m , then

$$[T_C]_{S',S} = C \quad (6)$$

EXAMPLE 1 | Matrix for a Linear Transformation

Let $T : P_1 \rightarrow P_2$ be defined by $T(p(x)) = xp(x)$. Find the matrix for T with respect to the standard bases $B = \{\mathbf{u}_1, \mathbf{u}_2\} = \{1, x\}$ and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{1, x, x^2\}$.

SOLUTION: Compute:

$$T(\mathbf{u}_1) = T(1) = x, \quad T(\mathbf{u}_2) = T(x) = x^2$$

Their coordinate vectors relative to B' are

$$[T(\mathbf{u}_1)]_{B'} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad [T(\mathbf{u}_2)]_{B'} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

so by Formula (4),

$$[T]_{B',B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

EXAMPLE 2 | The Three-Step Procedure

Use the matrix from Example 1 to compute $T(a + bx) = x(a + bx) = ax + bx^2$.

SOLUTION: Step 1. $[\mathbf{x}]_B = \begin{bmatrix} a \\ b \end{bmatrix}$.

Step 2. $[T]_{B',B}[\mathbf{x}]_B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ a \\ b \end{bmatrix} = [T(\mathbf{x})]_{B'}$.

Step 3. $T(a + bx) = 0 \cdot 1 + a \cdot x + b \cdot x^2 = ax + bx^2$. ✓

EXAMPLE 3 | Matrix for a Linear Transformation

Let $T : R^2 \rightarrow R^3$ be defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 \\ -5x_1 + 13x_2 \\ -7x_1 + 16x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -5 & 13 \\ -7 & 16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Find $[T]_{B',B}$ relative to the bases $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ for R^2 and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3 , where

$$\mathbf{u}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}; \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

SOLUTION: First compute:

$$T(\mathbf{u}_1) = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \quad T(\mathbf{u}_2) = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

Expressing these as linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ (verify):

$$T(\mathbf{u}_1) = \mathbf{v}_1 - 2\mathbf{v}_3, \quad T(\mathbf{u}_2) = 3\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3$$

so

$$[T(\mathbf{u}_1)]_{B'} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \quad [T(\mathbf{u}_2)]_{B'} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$$

and

$$[T]_{B',B} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ -2 & -1 \end{bmatrix}$$

Remark. A fixed linear transformation has *multiple* matrix representations depending on the bases. Here

$$[T]_{S',S} = \begin{bmatrix} 0 & 1 \\ -5 & 13 \\ -7 & 16 \end{bmatrix} \quad \text{and} \quad [T]_{B',B} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ -2 & -1 \end{bmatrix}$$

both represent T : the first relative to the standard bases S and S' , the second relative to B and B' .

Matrices of Linear Operators

When $V = W$ (so $T : V \rightarrow V$ is a linear *operator*), we take $B = B'$. The resulting matrix is called the **matrix for T relative to the basis B** , denoted $[T]_B$. Formulas (4) and (5) specialize to

$$\boxed{[T]_B = \left[[T(\mathbf{u}_1)]_B \mid [T(\mathbf{u}_2)]_B \mid \cdots \mid [T(\mathbf{u}_n)]_B \right]} \quad (7)$$

$$\boxed{[T]_B[\mathbf{x}]_B = [T(\mathbf{x})]_B} \quad (8)$$

For matrix operators $T_A : R^n \rightarrow R^n$ with B the standard basis, Formula (7) reduces to

$$[T_A]_B = A \quad (9)$$

Matrices of Identity Operators

EXAMPLE 4 | Matrices of Identity Operators

If $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is a basis for V and $I : V \rightarrow V$ is the identity operator, then $I(\mathbf{u}_i) = \mathbf{u}_i$, so $[I(\mathbf{u}_i)]_B = \mathbf{e}_i$. Thus $[I]_B = I$ (the $n \times n$ identity matrix) for every basis B .

EXAMPLE 5 | Linear Operator on P_2

Let $T : P_2 \rightarrow P_2$ be the linear operator defined by $T(p(x)) = p(3x - 5)$, that is, $T(c_0 + c_1x + c_2x^2) = c_0 + c_1(3x - 5) + c_2(3x - 5)^2$.

- (a) Find $[T]_B$ relative to the basis $B = \{1, x, x^2\}$.
- (b) Use the indirect procedure to compute $T(1 + 2x + 3x^2)$.
- (c) Check part (b) by direct computation.

SOLUTION (a). Compute the images of the basis vectors:

$$T(1) = 1, \quad T(x) = 3x - 5, \quad T(x^2) = (3x - 5)^2 = 9x^2 - 30x + 25$$

Their coordinate vectors relative to B :

$$[T(1)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [T(x)]_B = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix}, \quad [T(x^2)]_B = \begin{bmatrix} 25 \\ -30 \\ 9 \end{bmatrix}$$

so

$$[T]_B = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix}$$

SOLUTION (b). Let $\mathbf{p} = 1 + 2x + 3x^2$.

Step 1. $[\mathbf{p}]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Step 2. $[T]_B[\mathbf{p}]_B = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 66 \\ -84 \\ 27 \end{bmatrix} = [T(\mathbf{p})]_B$.

Step 3. $T(1 + 2x + 3x^2) = 66 - 84x + 27x^2$.

SOLUTION (c). Direct computation:

$$\begin{aligned} T(1 + 2x + 3x^2) &= 1 + 2(3x - 5) + 3(3x - 5)^2 \\ &= 1 + 6x - 10 + 3(9x^2 - 30x + 25) = 66 - 84x + 27x^2 \quad \checkmark \end{aligned}$$

Matrices of Compositions and Inverse Transformations

Theorem 8.4.1

If $T_1 : U \rightarrow V$ and $T_2 : V \rightarrow W$ are linear transformations with bases B, B'', B' for U, V, W respectively, then

$$[T_2 \circ T_1]_{B',B} = [T_2]_{B',B''}[T_1]_{B'',B} \quad (10)$$

Theorem 8.4.2

If $T : V \rightarrow V$ is a linear operator and B is a basis for V , then the following are equivalent:

- (a) T is one-to-one.
- (b) $[T]_B$ is invertible.

Moreover, when these conditions hold,

$$[T^{-1}]_B = [T]_B^{-1} \quad (11)$$

Remark. Observe how the interior subscript B'' in Formula (10) "cancels," leaving only the bases for the domain and image. This cancellation extends to three transformations: if $T_1 : U \rightarrow V$, $T_2 : V \rightarrow W$, $T_3 : W \rightarrow Y$, then

$$[T_3 \circ T_2 \circ T_1]_{B',B} = [T_3]_{B',B'''}[T_2]_{B''',B''}[T_1]_{B'',B} \quad (12)$$

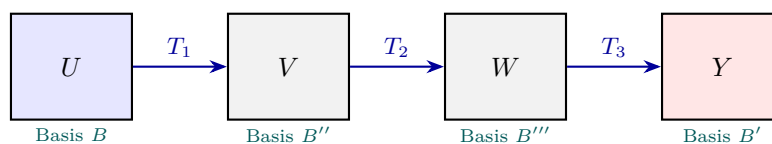


FIGURE 8.4.6

EXAMPLE 6 | Composition

Let $T_1 : P_1 \rightarrow P_2$ be defined by $T_1(p(x)) = xp(x)$ and $T_2 : P_2 \rightarrow P_2$ by $T_2(p(x)) = p(3x - 5)$. The composition $(T_2 \circ T_1) : P_1 \rightarrow P_2$ is

$$(T_2 \circ T_1)(p(x)) = T_2(xp(x)) = (3x - 5)p(3x - 5)$$

so for $p(x) = c_0 + c_1x$,

$$(T_2 \circ T_1)(c_0 + c_1x) = c_0(3x - 5) + c_1(3x - 5)^2 \quad (13)$$

In Theorem 8.4.1, P_2 plays both the role of V and W , so $B'' = B' = \{1, x, x^2\}$. Formula (10) simplifies to

$$[T_2 \circ T_1]_{B',B} = [T_2]_{B'}[T_1]_{B',B} \quad (14)$$

Using $[T_1]_{B',B}$ from Example 1 and $[T_2]_{B'}$ from Example 5:

$$[T_2 \circ T_1]_{B',B} = \begin{bmatrix} 1 & -5 & 25 \\ 0 & 3 & -30 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 25 \\ 3 & -30 \\ 0 & 9 \end{bmatrix} \quad (15)$$

Verification. From (13): $(T_2 \circ T_1)(1) = 3x - 5$ and $(T_2 \circ T_1)(x) = (3x - 5)^2 = 9x^2 - 30x + 25$. Their coordinate vectors w.r.t. B' are $[-5; 3; 0]$ and $[25; -30; 9]$, giving $[T_2 \circ T_1]_{B',B} = \begin{bmatrix} -5 & 25 \\ 3 & -30 \\ 0 & 9 \end{bmatrix}$, which agrees with (15). ✓

Exercise Set 8.4

In Exercises 1–8, find the matrix for the linear transformation relative to the indicated bases.

1. Let $T : P_2 \rightarrow P_3$ be defined by $T(p(x)) = xp(x)$.
 - a. Find the matrix for T relative to the standard bases $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \{1, x, x^2\}$ and $B' = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} = \{1, x, x^2, x^3\}$.
 - b. Verify that $[T]_{B',B}$ satisfies Formula (5) for every $\mathbf{x} = c_0 + c_1x + c_2x^2$ in P_2 .
3. Let $T : P_2 \rightarrow P_2$ be the linear operator defined by $T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x - 1) + a_2(x - 1)^2$.
 - a. Find the matrix for T relative to the standard basis $B = \{1, x, x^2\}$.
 - b. Verify that $[T]_B$ satisfies Formula (8) for every $\mathbf{x} = a_0 + a_1x + a_2x^2$ in P_2 .
4. Let $T : R^2 \rightarrow R^2$ be the linear operator defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$, and let $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ where $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.
 - a. Find $[T]_B$.
 - b. Verify that $[T]_B$ satisfies Formula (8) for every vector $\mathbf{x}$ in R^2 .
7. Let $T : P_2 \rightarrow P_2$ be the linear operator $T(p(x)) = p(2x + 1)$, that is, $T(c_0 + c_1x + c_2x^2) = c_0 + c_1(2x + 1) + c_2(2x + 1)^2$.
 - a. Find $[T]_B$ with respect to the basis $B = \{1, x, x^2\}$.
 - b. Use the three-step procedure to compute $T(2 - 3x + 4x^2)$.
 - c. Check the result in (b) by computing $T(2 - 3x + 4x^2)$ directly.

In Exercises 9–11, the matrix A for T relative to a given basis is provided; find T explicitly.

9. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$ be the matrix for $T : R^2 \rightarrow R^2$ relative to $B = \{\mathbf{v}_1, \mathbf{v}_2\}$.
 - a. Find $[T(\mathbf{v}_1)]_B$ and $[T(\mathbf{v}_2)]_B$.
 - b. Find $T(\mathbf{v}_1)$ and $T(\mathbf{v}_2)$.
 - c. Find a formula for $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$.

- d. Use the formula in (c) to compute $T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$.
11. Let $A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 0 & 5 \\ 6 & -2 & 4 \end{bmatrix}$ be the matrix for $T : P_2 \rightarrow P_2$ with respect to the basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = 3x + 3x^2$, $\mathbf{v}_2 = -1 + 3x + 2x^2$, $\mathbf{v}_3 = 3 + 7x + 2x^2$.
- Find $[T(\mathbf{v}_1)]_B$, $[T(\mathbf{v}_2)]_B$, and $[T(\mathbf{v}_3)]_B$.
 - Find $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$, and $T(\mathbf{v}_3)$.
 - Find a formula for $T(a_0 + a_1x + a_2x^2)$.
 - Use the formula in (c) to compute $T(1 + x^2)$.
12. Let $T_1 : P_1 \rightarrow P_2$ be defined by $T_1(p(x)) = xp(x)$ and let $T_2 : P_2 \rightarrow P_2$ be defined by $T_2(p(x)) = p(2x + 1)$. Let $B = \{1, x\}$ and $B' = \{1, x, x^2\}$ be the standard bases for P_1 and P_2 .
- Find $[T_2 \circ T_1]_{B', B}$, $[T_2]_{B'}$, and $[T_1]_{B', B}$.
 - State a formula relating the matrices in part (a).
 - Verify that the matrices in part (a) satisfy the formula in (b).
14. Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be a basis for a vector space V . Find the matrix with respect to B for the linear operator $T : V \rightarrow V$ defined by $T(\mathbf{v}_1) = \mathbf{v}_2$, $T(\mathbf{v}_2) = \mathbf{v}_3$, $T(\mathbf{v}_3) = \mathbf{v}_4$, $T(\mathbf{v}_4) = \mathbf{v}_1$.
17. (*Calculus required*) Let $D : P_2 \rightarrow P_2$ be the differentiation operator $D(\mathbf{p}) = p'(x)$.
- Find the matrix for D relative to the basis $B = \{p_1, p_2, p_3\}$ where $p_1 = 1$, $p_2 = x$, $p_3 = x^2$.
 - Use the matrix in (a) to compute $D(6 - 6x + 24x^2)$.
21. In each part, fill in the missing subscript.
- $[T_2 \circ T_1]_{B', B} = [T_2]_{\quad, B} [T_1]_{B'', B}$
 - $[T_3 \circ T_2 \circ T_1]_{B', B} = [T_3]_{\quad, B} [T_2]_{B''', B''} [T_1]_{B'', B}$

Working with Proofs

22. Prove: if $T : V \rightarrow W$ is the zero transformation, then the matrix for T with respect to any bases for V and W is a zero matrix.
23. Prove: if B and B' are the standard bases for R^n and R^m respectively, then the matrix for a linear transformation $T : R^n \rightarrow R^m$ relative to B and B' is the standard matrix for T .